

Del(∇) Operator in Spherical

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8th March 2026

1 Curvilinear Coordinates

1.1 General coordinate system

If I have a coordinate system with basis u_1, u_2, u_3 respectively (similar to x, y, z of cartesian coordinate system) with length elements

$$dl_{u_1} = h_1 du_1$$

$$dl_{u_2} = h_2 du_2$$

$$dl_{u_3} = h_3 du_3$$

And therefore, the corresponding volume element of that coordinate system is

$$dV = dl_{u_1} dl_{u_2} dl_{u_3} = h_1 h_2 h_3 du_1 du_2 du_3$$

1.2 Spherical Coordinate system

For spherical Coordinate system, the basis are r, θ and ϕ . The corresponding length elements are

$$dl_r = dr$$

$$dl_\theta = r d\theta$$

$$dl_\phi = r \sin(\theta) d\phi$$

And therefore, the corresponding volume element of that coordinate system is

$$dV = dl_r dl_\theta dl_\phi = r^2 \sin(\theta) dr d\theta d\phi$$

2 Different Del Operator uses

2.1 Gradient

The gradient function acts on scalar fields to give a vector field for the direction of increase in the scalar field. Lets say I have a scalar field V .

Gradient in Cartesian is given by

$$\nabla V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}$$

For our general coordinate system

$$\nabla V = \frac{1}{h_1} \frac{\partial V}{\partial u_1} \hat{u}_1 + \frac{1}{h_2} \frac{\partial V}{\partial u_2} \hat{u}_2 + \frac{1}{h_3} \frac{\partial V}{\partial u_3} \hat{u}_3$$

For Spherical Coordinate system

$$\nabla V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} + \frac{1}{r \sin(\theta)} \frac{\partial V}{\partial \phi} \hat{\phi}$$

2.2 Divergence

The divergence operator acts on vector fields and gives a scalar field that gives a scalar value to each point in space corresponding how good of a source or sink of vector fields that point is. Let $\vec{V}$ denote a vector field in space.

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$$

Divergence in Cartesian is given by

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

If I can write $\vec{V}$ in any general coordinates instead of cartesian, I can write the divergence operator in terms of those general coordinates

$$\nabla \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left(\frac{\partial}{\partial u_1} (h_2 h_3 V_{u_1}) + \frac{\partial}{\partial u_2} (h_3 h_1 V_{u_2}) + \frac{\partial}{\partial u_3} (h_1 h_2 V_{u_3}) \right)$$

For Spherical Coordinates

$$\nabla \cdot \vec{V} = \frac{1}{r^2 \sin(\theta)} \left(\frac{\partial}{\partial r} (r^2 \sin(\theta) V_r) + \frac{\partial}{\partial \theta} (r \sin(\theta) V_\theta) + \frac{\partial}{\partial \phi} (r V_\phi) \right)$$

That simplifies to

$$\nabla \cdot \vec{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) V_\theta) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} (V_\phi)$$

2.3 Curl

The Curl operator acts of Vector fields and gives another vector field that contains information about how much our original vector field rotates around points in space. The magnitude of the curl is maximum at points where the vector

field rotates most tightly around a point. The direction of curl at a point is given by the Right hand thumb rule of the vector field at that point. Let $\vec{V}$ be a vector field in cartesian space given by

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$$

Curl in Cartesian is given by

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

For General coordinate system,

$$\nabla \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{u}_1 & h_2 \hat{u}_2 & h_3 \hat{u}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 V_{u_1} & h_2 V_{u_2} & h_3 V_{u_3} \end{vmatrix}$$

For Spherical Coordinate system, that would become

$$\nabla \times \vec{V} = \frac{1}{r^2 \sin(\theta)} \begin{vmatrix} \hat{r} & r \hat{\theta} & r \sin(\theta) \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ V_r & r V_\theta & r \sin(\theta) V_\phi \end{vmatrix}$$

2.4 Laplacian

The Laplacian written as ∇^2 is the divergence of the gradient of a scalar function. So, its output is a Scalar function. If V is a scalar function

$$\nabla^2 V = \nabla \cdot (\nabla V)$$

For Cartesian, that's just

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

To get the expression for Laplacian in a general coordinate system, we write expression for divergence and instead of V_{u_1} , V_{u_2} and V_{u_3} , we write the corresponding terms of the gradient of that system. That is

$$\nabla^2 V = \frac{1}{h_1 h_2 h_3} \left(\frac{\partial}{\partial u_1} h_2 h_3 \left(\frac{1}{h_1} \frac{\partial V}{\partial u_1} \right) + \frac{\partial}{\partial u_2} h_3 h_1 \left(\frac{1}{h_2} \frac{\partial V}{\partial u_2} \right) + \frac{\partial}{\partial u_3} h_1 h_2 \left(\frac{1}{h_3} \frac{\partial V}{\partial u_3} \right) \right)$$

For spherical,

$$\nabla^2 V = \frac{1}{r^2 \sin(\theta)} \left(\frac{\partial}{\partial r} \left(r^2 \sin(\theta) \frac{\partial V}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(r \sin(\theta) \frac{1}{r} \frac{\partial V}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(r \frac{1}{r \sin(\theta)} \frac{\partial V}{\partial \phi} \right) \right)$$

that simplifies to

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 V}{\partial \phi^2}$$